CE / CS / SE 3354 · SOFTWARE ENGINEERING · FALL 2026

Project Deliverable P1

Proposal and Requirements — Study Room Sniper

Weight 4% · Due Sunday 18 October 2026, 11:59 PM Central · Submit one PDF per team on Canvas

Field	Your entry
Project title	Study Room Sniper — availability alerts for McDermott Library study rooms
Team number and section	Team 4, Section 003
Team members (name · net ID)	Alexander Eng - dal084253, Arshan Khimani – axk230294, Lee Baillard - leb240001, Mayand Dhami, Henry McGahan, Isaam Memon
Repository URL	github.com/PaperBagTrick/Study-Room-Sniper
Document version	0.1
Date	18 October 2026

0 Revision History and Contribution Record

Version	Date	Author	Summary of change
0.1	16 October 2026	Team 4 (Alexander Eng, Arshan Khimani, Lee Baillard, Mayand Dhami, Henry McGahan, Isaam Memon)	Initial baseline for P1

0.1 Individual contribution record

Team member	What this person produced for P1	Evidence in the repository
Alexander Eng	Wrote and assembled the P1 document from the course template: §1.1–1.4, §2 and §5; maintained the revision history, contribution record and baseline declaration; ran the change-control process for v0.1.	Commit on main ("P1 baseline v0.1"); issue #1 "P1 document assembly"; PR #2 "P1 baseline review".
Arshan Khimani	Wrote §2.5 Assumptions (A1–A6), §2.6 Feasibility summary and the technical half of §4.2; set up the repository and branch/commit conventions (§4.4).	Commits touching §2.5–2.6 in the range above; issue #3 "A1 availability-read feasibility"; repo settings documented in /docs/tooling.md.
Lee Baillard	Wrote the stakeholder elicitation plan and questions: §2.2 stakeholder table, §4.1 elicitation role, Q1–Q4 and Q6 in §5, and the 23 Oct contact milestones in §4.2.	Issue #5 "Elicitation plan Q1–Q6"; /docs/elicitation/plan.md in commit; PR #6 reviewed by Mayand.
Mayand Dhami	Wrote quality requirements QR-01–QR-06 (§3.2), the accessibility constraint QR-04/WCAG references, the survey and usability-test plan for Q5 (25 Oct) and QR-03/QR-01 verification.	Commit "Add QR-01–QR-06"; issue #8 "QR-04 WCAG AA acceptance method"; /docs/elicitation/survey-draft.md.
Henry McGahan	Wrote §3.4 use cases UC-01–UC-04 (expanded form with flows and postconditions) and §3.5 traceability matrix, including the "what the matrix revealed" notes.	Commit "Add UC-01–UC-04 and traceability matrix"; PR #9 reviewed by Isaam; issue #10 "Matrix orphan check".

Isaam Memon	Wrote the risk register R1–R5 and the integration/verification steps in §4.2 (4 Nov–8 Nov); checked free-tier limits for email/push (§2.6, R5).	Commit "Add §4.3 risks and integration plan"; issue #11 "Free-tier limit check"; PR #12 "P2 integration checklist".

1 Introduction

1.1 Purpose of this document and of the system

This document is the P1 baseline for Team 4's CS/SE 3354 project. It states the problem that Study Room Sniper addresses, the scope the team has committed to, the requirements the system must meet, and the plan for reaching P2. It is written for the course instructor and TA, who will assess it, and for the six team members, who will build from it. The requirements in section 3 follow the style of ISO/IEC/IEEE 29148 [R1].

Study Room Sniper addresses a problem for students who need a study room at Eugene McDermott Library. Rooms are reserved through the library's online system [R2], and when every room for the time a student needs is taken, the only way to learn that one has opened (a reservation is cancelled or lapses after the 15-minute hold [R3]) is to keep refreshing the booking page. Study Room Sniper lets a currently enrolled UTD student describe the room they need and alerts them as soon as a matching room becomes bookable. It does not book the room for them.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Watches for the 9 group study rooms and 8 smart rooms listed for McDermott Library [R2]	**Booking, extending or cancelling any room on a student's behalf.** It would need the student's library credentials or an agreement with the library, and could let one student crowd out others.
Read-only, periodic checking of room availability in the library's reservation system	**Rooms outside McDermott Library, and the One Button Studio [R2].** Their booking rules are assumed to differ, and the team cannot verify them before P2.
Alerts by email and by browser push notification, each with a link to the library's booking page	**SMS alerts.** Messaging has a per-message cost, and the team has no budget (CON-03).
Sign-in with UTD NetID; each student sees and manages only their own watches	**Native mobile apps.** A responsive web interface reaches the same phones at far lower cost.
A basic operator view showing whether the checking of library availability is healthy	**Predicting when rooms are likely to free up or recommending rooms.** It needs historical data that the team does not have.

Why the exclusions were excluded:

Booking on a student's behalf was excluded first and most firmly: it depends on permissions the team does not have and on credentials we do not want to hold, and an alerting tool is useful without it. The other exclusions were left out because each adds cost, data or verification effort that the six-person team cannot absorb before P2 on 8 November, and none is needed for a student to learn that a room has opened.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
Study Room Sniper	The system specified in this document. Despite the name, it only alerts; it never reserves a room (CON-02).

Watch	A student's saved request to be alerted: a date, a time window and a minimum group size. Its status is Active, Expired, or Cancelled.
Slot	A period during which one study room can be reserved, as the library reservation system presents it.
Bookable	**Two meanings, both in use.** (1) Free: the library system shows the slot as unreserved. (2) Reservable by this student now: free, and also within the 7-day booking horizon [R2] and the student's own limits (one active reservation per 24 hours, at most four hours [R3]). A slot can be free but not bookable for a given student. Resolved in Q4; until then "bookable" in this document means free.
Match	A bookable slot that satisfies an Active watch: same date, entirely inside the watch's time window, in a room whose listed capacity fits the group size.
Read cycle (polling)	One pass in which the system reads current availability for every in-scope room from the library reservation system. Reads are read-only requests.
Alert	A message (email, optionally browser push) telling a student that a match exists.
Library Reservation System (LRS)	The library's online study-room booking service (UTD LibCal, [R2]). External: the team can read from it but does not build or control it.
NetID; SSO	NetID is a student's UTD login identifier. SSO (single sign-on) lets a student sign in with it without creating a separate account.
Operator	A team member responsible for running the service and checking that it is healthy.
MoSCoW	Priority scheme used in section 3: MUST, SHOULD, COULD, WON'T.
FR, QR, CON	Identifier prefixes for functional, quality, and constraint requirements.
WCAG	Web Content Accessibility Guidelines, the accessibility standard cited in [R5].

1.4 References

[R1] ISO/IEC/IEEE 29148:2018. Systems and Software Engineering — Life cycle processes — Requirements engineering.

[R2] Eugene McDermott Library, "Study Rooms" (UTD LibCal), https://libcal.utdallas.edu/ — room counts, hours, 7-day booking horizon. Accessed 16 October 2026.

[R3] Eugene McDermott Library, "Reserve a Study Room," https://library.utdallas.edu/about-the-library/reserve-a-study-room — enrolled students only; four-hour maximum; one active reservation per 24 hours; 15-minute hold. Accessed 16 October 2026.

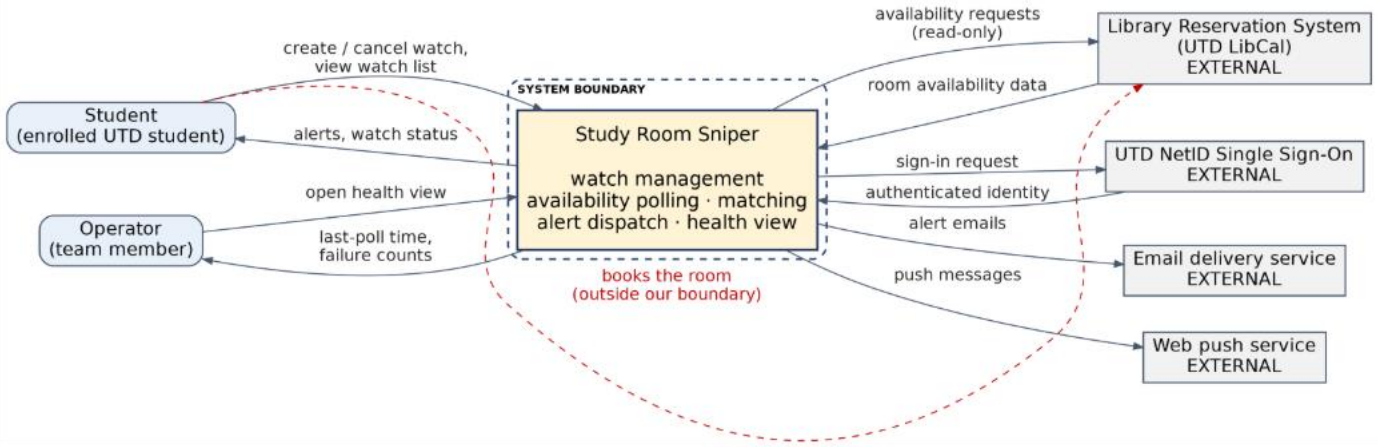
[R4] Eugene McDermott Library, "Space Availability — Study Rooms" policies, https://libcal.utdallas.edu/reserve/spaces/study-rooms — occupancy limits, floors and closing times. Accessed 16 October 2026.

[R5] W3C, Web Content Accessibility Guidelines (WCAG) 2.2, https://www.w3.org/TR/WCAG22/.

[R6] CS/SE 3354 Fall 2026 syllabus and Project Deliverable P1 template (course documents on Canvas).

2 Overall Description

2.1 Product perspective and context



Everything inside the dashed boundary is built and run by the team: the watch list, the periodic reading of availability, the matching of availability against watches, the sending of alerts and the operator health view. Everything the system calls on but does not build sits outside it: the library reservation system (the source of truth for availability, which we can read but not change), UTD NetID sign-on (which identifies students), and the email and push services that carry alerts. The boundary stops short of booking on purpose. The student, not the system, makes the reservation on the library's own site, so the system never holds library credentials and the library keeps control of who gets a room (CON-02).

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Students who want a group study room	Primary users. Create watches and receive alerts.	To learn quickly when a room matching their need opens, with little effort to set up, and an alert that arrives in time to act on.	**Assumption.** Based on the team's position as UTD students and on the booking rules in [R2] and [R3]. No interviews yet; a survey of at least 10 students is planned for 25 October (4.2, Q5).
Students who use assistive technology or have other access needs	Primary users with specific constraints.	A watch list and alerts that work by keyboard and screen reader, with nothing conveyed by color or sound alone.	**Assumption**, taken from WCAG 2.2 [R5]. Not yet consulted (Q6).
McDermott Library reservations staff	Operate the reservation system we read from. Never use ours.	That our traffic does not degrade their service, and that their rules for fair access to rooms (for example one active reservation per 24 hours [R3]) are not undermined.	**Assumption**, inferred from [R2]–[R4]. No contact yet (Q1, Q2).
UTD Office of Information Technology (OIT)	Operates NetID sign-on and campus IT policy. Never uses ours.	That only approved services use NetID sign-on and that student data is limited to what the service needs.	**Assumption.** No contact yet (Q3, Q6).
Project team (Alexander Eng, Arshan Khimani, Lee Baillard, Mayand Dhami, Henry McGahan, Isaam Memon)	Builders and operators.	Free hosting, a service that is easy to monitor, and requirements stable enough to build against by P2.	Direct knowledge: the team's own needs and the deadlines in [R6].

2.3 User characteristics

Students. The users are currently enrolled UTD students, the only people allowed to reserve study rooms [R3]. They are assumed to be comfortable with web apps, to hold a NetID and to read most messages on a phone. They cannot be assumed to know how the system works or to read documentation, and they usually arrive in a hurry: either planning a session within the 7-day booking window [R2] or already in the library without a room. Alerts therefore have to be readable at a glance on a phone, and creating a watch has to take well under two minutes (QR-03).

Students with access needs. Some students use screen readers, keyboard-only navigation, magnification or reduced motion. The system has to work for them in the same interface rather than a separate version (QR-04). An alert that relies on sound or color alone would fail them.

Operators. The team members who run the service. They are comfortable with logs and web consoles and need only a simple view of when library availability was last read successfully and how many recent reads failed.

2.4 Constraints

CON-01 Sign-in uses UTD NetID single sign-on only. Source: assumed OIT requirement for campus services; to be confirmed (Q3).

CON-02 The system only reads from the library reservation system and never creates, changes or cancels a reservation. Source: library reservation policy, which ties each reservation to one enrolled student [R3], and the team's decision not to hold student credentials. To be confirmed with the library (Q2).

CON-03 The system is built and run using only services that cost the team nothing. Source: team decision; the team has no budget.

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	The library reservation system lets a program read room availability without a student's credentials, for example from the public availability pages on libcal.utdallas.edu [R2].	FR-03 and FR-04 cannot be met as written. The team would have to hold student credentials, which CON-02 forbids, or fall back to demonstrating the system against recorded sample data.
A2	The library accepts automated, read-only checking at a modest rate: no more than 20 requests per minute in total (QR-05).	The library may block the service. The team would have to lower the read rate, which weakens QR-01, or stop.
A3	UTD lets a student-built service use NetID sign-on, and releases at least a NetID and an email address.	FR-02 cannot be met. The team would raise a change request to CON-01 and use email-verified accounts, losing the assurance that users are UTD students.
A4	The set of rooms stays small: 17 group study and smart rooms today [R2], and no more than 20 in future.	More rooms would strain the request budget in QR-05: at one request per room, 20 requests per minute covers at most 40 rooms every two minutes, so beyond that FR-03 could not be met.
A5	Students see an email or push alert on their phone within a few minutes while they are looking for a room.	Alerts arrive on time but go unseen and the benefit shrinks. The team would revisit alert channels and the SMS exclusion.
A6	Each room's capacity can be obtained from the library's pages [R4].	Matching on group size (FR-01, FR-04) is impossible. The group-size input would be dropped through a change request.

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Likely feasible; confirmed only once the spike on A1 is done	The work is periodic reading, comparison against saved watches and message sending, with no novel algorithm. The one real unknown is whether availability can be read without credentials (A1). A spike is planned for 21 October with findings to be recorded under /docs/spikes in the repository (4.2). No spike has been done yet.

Operational	Feasible, subject to the library not objecting (Q2)	Students already reserve through the library system [R2]; Study Room Sniper adds an alert step and changes nothing in the library's process. Running it needs only a periodic health check by an operator (UC-04).
Schedule	Feasible for the P2 baseline; tight	Section 4.2 has eight milestones between 18 October and 8 November for the six-person team, with the riskiest items (spike, library and OIT contact) first. P1 asks for requirements, not code [R6].
Economic / resource	Feasible	No budget is needed if no paid service is used (CON-03). Free tiers for email and push delivery are assumed sufficient for the expected volume; the team will check their limits in the 21 October spike.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one "shall" sentence	Priority	Source	Acceptance criterion
FR-01	The system shall let a signed-in student create a watch by giving a date within the next 7 days, a time window and a minimum group size.	MUST	Students (2.2); [R2]	A tester creates a watch with each input and sees it in the watch list with the values entered; a date more than 7 days ahead is refused with a message.
FR-02	The system shall require a student to sign in with UTD NetID before creating, viewing or cancelling a watch.	MUST	CON-01	A tester tries each of the three functions while signed out and is sent to NetID sign-in each time; after signing in the function is available.
FR-03	The system shall read the availability of every in-scope room from the library reservation system at least once every 2 minutes while the library is open.	MUST	Students (2.2); [R2]	Against a stub library system that is always reachable, a tester reviews the request log for a 60-minute period and finds no room with a gap above 120 seconds between reads.
FR-04	The system shall recognize a match when a bookable slot that satisfies an Active watch appears in the availability data.	MUST	Students (2.2)	Using the stub, a tester releases one slot that fits a watch and one that does not (wrong time); within one read cycle the first is a match and the second is not.
FR-05	The system shall confirm that a slot is still bookable immediately before sending an alert for it.	SHOULD	Students (2.2)	Using the stub, a tester makes a matching slot disappear between the read and the send; no alert is sent for it.
FR-06	The system shall email the student an alert covering every match found in a read cycle.	MUST	Students (2.2)	Using the stub, a tester releases a matching slot and finds exactly one alert in the test inbox for the student who owns the watch.
FR-07	The system shall include in each alert the room name, the slot's start and end times and a link to the library's booking page.	MUST	Students (2.2); [R2]	A tester opens a received alert, checks that all three items are present and that the link opens the library booking page.
FR-08	The system shall let a student choose browser push as an additional alert channel for a watch.	SHOULD	Students (2.2)	A tester enables push on one watch and not on another and releases a match for each; the first produces an email and a push message, the second only an email.
FR-09	The system shall let a student cancel an Active watch.	MUST	Students (2.2)	A tester cancels an Active watch, then releases a slot that would have matched; no

				alert is sent and the watch shows as Cancelled.
FR-10	The system shall show a student a list of their own watches with the status of each (Active, Expired or Cancelled).	MUST	Students (2.2)	Using two accounts, a tester creates, cancels and expires watches; each account sees only its own watches, each with the correct status.
FR-11	The system shall mark a watch Expired when the end of its time window has passed and stop evaluating it.	MUST	Students; Library staff (2.2)	Using a test clock, a tester moves time past a watch's window end; within 5 minutes the watch is Expired and a later matching slot produces no alert.
FR-12	The system shall send at most one alert per slot for each watch.	MUST	Students (2.2)	With a slot that stays free across 10 read cycles, a tester receives exactly one alert for it.
FR-13	The system shall allow a student no more than 5 Active watches at one time.	SHOULD	Library staff; Team (2.2)	A tester creates 5 watches and tries a sixth, which is refused with a message; after cancelling one, a new watch can be created.
FR-14	The system shall show an operator the time of the last successful read of library availability and the number of failed reads in the last hour.	COULD	Team (2.2)	A tester stops the stub for 10 minutes; the operator view shows a last-read time that stops advancing and a failure count above zero, and both recover after the stub restarts.

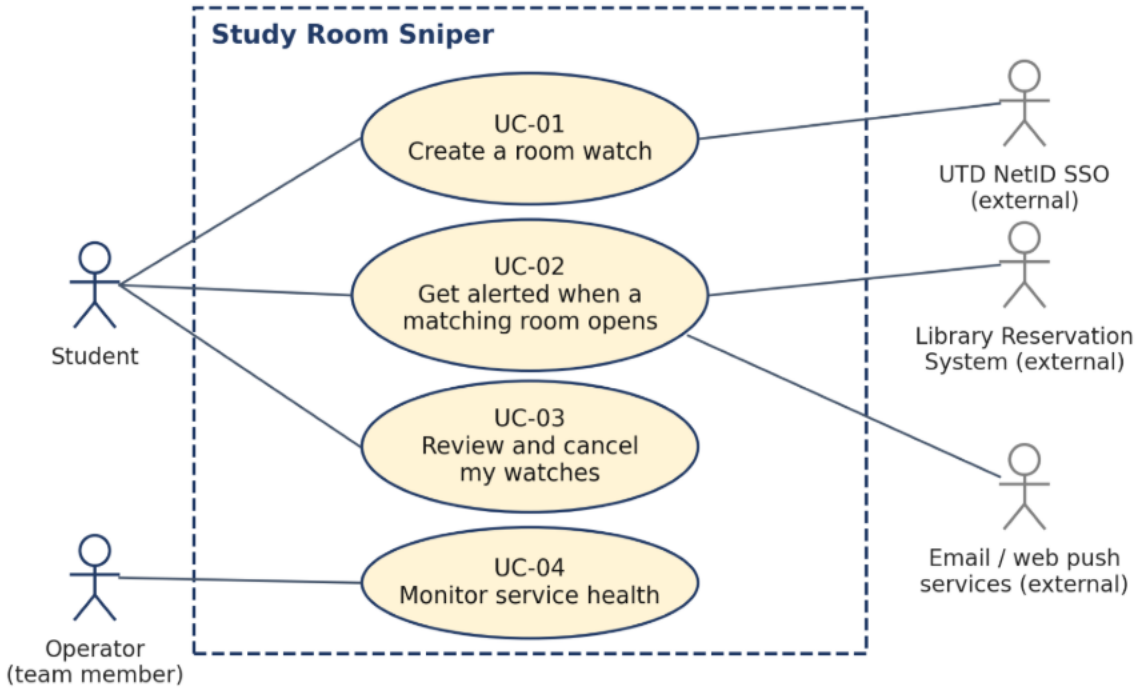
3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	Ninety-five percent of alerts shall be delivered to the student's email service within 180 seconds of the slot becoming bookable in the library system, measured under a load of up to 200 Active watches.	MUST	Students (2.2)	Scripted releases against the stub: compare each release time with the delivery time in the logs over 100 releases; pass if at least 95 are within 180 seconds.
QR-02	The service shall be available at least 99% of the time during library opening hours, measured over a continuous 14-day trial.	SHOULD	Students; Team (2.2)	An automated check every 5 minutes; availability is successful checks divided by all checks during opening hours, and must be at least 99%.
QR-03	At least 90% of first-time users shall create a watch in under 90 seconds without help, measured on a phone.	MUST	Students (2.2)	A moderated test with at least 10 UTD students, timed from the first screen to the watch appearing in the list; pass if at least 9 of 10 succeed.
QR-04	Every screen used in UC-01 and UC-03 shall meet WCAG 2.2 Level AA [R5], with zero violations rated serious or critical.	MUST	Students with access needs (2.2); [R5]	An automated accessibility scan of each screen, plus a keyboard-only and a screen-reader walk-through of UC-01 and UC-03 in which every step can be completed.
QR-05	The system shall send no more than 20 requests per minute in total to the library reservation system, however many watches are Active.	MUST	Library staff (2.2); A2	Count the requests in the request log over a 24-hour run with 200 Active watches; pass if no minute exceeds 20.
QR-06	The system shall store no personal data about a student other than their NetID, email address and watches.	MUST	OIT; Students (2.2)	Review a full export of the stored data for three test accounts and confirm that nothing else about the student is present.

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall use UTD NetID single sign-on as its only means of student sign-in.	OIT (assumed, Q3)	The team has no other way to show that a user is a UTD student, and cannot issue accounts the library or OIT would trust. Confirmed or relaxed through Q3.
CON-02	The system shall only read from the library reservation system and shall never create, change or cancel a reservation.	[R3]; team decision	Acting for a student needs that student's credentials and the library's consent. Neither exists, and holding credentials is a security risk the team will not take.
CON-03	The system shall be built and run using only services that cost the team nothing.	Team (no budget)	The team has no funds, and a paid service would make the project impossible to keep running.

3.4 Initial use cases



Four use cases, each finishing something a student or operator came to do. Booking the room happens on the library's own site and is outside the system (2.1). Signing in is a step inside UC-01 and UC-03, not a use case of its own.

Expanded use case UC-01 — Create a room watch

Field	Content
Identifier and name	UC-01 Create a room watch
Primary actor	Student
Supporting actors	UTD NetID single sign-on (external)
Preconditions — what must be true before it starts	The student is currently enrolled and has a NetID. The service is running.

Trigger — what starts it	The student finds no room they can book for the time they need and decides to be told when one opens.
Main success scenario — numbered steps	1. The student opens Study Room Sniper. 2. The system asks the student to sign in; the student signs in with NetID. 3. The student chooses to create a new watch. 4. The student enters a date, a time window and a minimum group size. 5. The student confirms. Email is the alert channel unless the student also enables browser push. 6. The system checks the entries and the student's number of Active watches. 7. The system saves the watch as Active and shows it in the student's watch list.
Alternative and exception flows	2a. Sign-in fails or is cancelled: the system says so and returns to step 2; no watch is created. 4a. The date is in the past or more than 7 days ahead [R2], or the window ends before it starts: the system explains and keeps the entries for correction. 5a. The browser refuses push permission: the system saves the watch with email only and says so. 6a. The student already has 5 Active watches (FR-13): the system refuses and offers the watch list so one can be cancelled (UC-03); the student then restarts at step 3. 6b. A matching slot is already bookable: the system shows it now with the booking link and asks whether to save the watch anyway; the answer decides whether step 7 happens.
Postconditions — what is true when it succeeds	On success an Active watch owned by the student exists and will be evaluated at the next read of library availability. On failure no watch is created and no other student's data is affected.
Requirements realized	FR-01, FR-02, FR-04, FR-08, FR-10, FR-13

Expanded use case UC-02 — Get alerted when a matching room opens

Field	Content
Identifier and name	UC-02 Get alerted when a matching room opens
Primary actor	Student
Supporting actors	Library Reservation System (external; source of availability). Email service and web push service (external; deliver the alert).
Preconditions — what must be true before it starts	The student has at least one Active watch whose window has not ended. The system is reading library availability.
Trigger — what starts it	A slot that fits an Active watch becomes bookable in the library reservation system.
Main success scenario — numbered steps	1. The system reads current availability for all in-scope rooms from the library reservation system. 2. The system finds a bookable slot that satisfies an Active watch (same date, inside the time window, room large enough). 3. The system confirms the slot is still bookable. 4. The system emails the student an alert giving the room, the start and end times and a link to the library's booking page, and sends a push message too if the student enabled it. 5. The system records that this slot has been alerted for this watch. 6. The student reads the alert and follows the link to the library's booking page, where the student can reserve the room (outside the system).
Alternative and exception flows	1a. The library system cannot be reached or returns data the system cannot read: it skips this cycle, records the failure and tries again at the next cycle; all watches stay Active. The failure shows in the operator view (UC-04). 2a. A watch's time window has ended: the system sets it to Expired and does not evaluate it (FR-11). 2b. Several slots match at once: they are sent together in a single alert. 2c. The slot was already alerted for this watch: it is skipped (FR-12). 3a. The slot is no longer bookable: no alert is sent and the watch stays Active. 4a. The email or push provider rejects the message: the system retries up to 3 times within 5 minutes, then records the failure in its log.

	6a. The slot is taken before the student acts: nothing more is sent; the watch stays Active and alerts on the next matching slot.
Postconditions — what is true when it succeeds	The student has been told about the slot and can reach the booking page. The slot is recorded as alerted for the watch, and the watch stays Active until it expires or is cancelled.
Requirements realised	FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-11, FR-12

3.5 Requirements to use case traceability

Requirement ID	Use case(s) that realise it	If blank — why, and what you will do
FR-01	UC-01	—
FR-02	UC-01, UC-03	—
FR-03	UC-02	—
FR-04	UC-01 (flow 6b), UC-02	—
FR-05	UC-02	—
FR-06	UC-02	—
FR-07	UC-02	—
FR-08	UC-01, UC-02	—
FR-09	UC-03	—
FR-10	UC-01, UC-03	—
FR-11	UC-02 (flow 2a), UC-03	—
FR-12	UC-02	—
FR-13	UC-01 (flow 6a)	—
FR-14	UC-04	—

What the matrix revealed, and what we changed because of it:

Every functional requirement reaches at least one use case and every use case has requirements behind it, so no row is blank and none of the four use cases is orphaned. Three things in the matrix are worth recording.

FR-14 is the only requirement behind UC-04. UC-04 exists only to realise it, and FR-14 is the one COULD requirement, so UC-04 is the first thing to cut if the schedule slips (risk R4).

FR-11 (expiry) has no student trigger. It happens without anyone acting, so on its own it would trace only to the watch list in UC-03. We added alternative flow 2a to UC-02 so that the moment of expiry sits inside a use case.

FR-13 (the 5-watch limit) reaches only a refusal flow in UC-01, which is why it is SHOULD rather than MUST.

Quality requirements (3.2) and constraints (3.3) are left out of the matrix on purpose: they apply across use cases rather than being realised by one.

4 Project Plan

4.1 Team roles and working agreement

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Alexander Eng	Document and change-control lead: owns this document, the revision history and the change-request process; submits P1.	Each owner decides within their area after hearing the others, with the evidence recorded in the issue. If another member objects, the team votes and a simple majority of the six decides. A tie, or a deadlock of 48 hours, goes to the instructor or TA.
Arshan Khimani	Technical lead: owns the library-availability spike, the feasibility evidence, the repository and tooling, and the architecture draft for P2.	Decides technical spikes, tooling and architecture after hearing the team; same voting and escalation rule as above.
Lee Baillard	Elicitation lead: owns contact with the library and OIT (Q1–Q4, Q6) and logs every reply in /docs/elicitation.	Decides what is asked of outside parties and in what words, after hearing the team; same voting and escalation rule.
Mayand Dhami	Quality and accessibility lead: owns the quality requirements (3.2), the WCAG check, and the student survey and usability test.	Decides quality thresholds and test methods after hearing the team; same voting and escalation rule.
Henry McGahan	Use case lead: owns the use case diagram, the expanded use cases, the traceability matrix, and the class and sequence diagrams for P2.	Decides use case boundaries and diagram content after hearing the team; same voting and escalation rule.
Isaam Memon	Risk and integration lead: owns the risk register, checks free-tier limits for email and push, and integrates the P2 document.	Decides risk ratings and the integration order after hearing the team; same voting and escalation rule.

4.2 Plan to P2

Date	What will be finished	Owner
Sun 18 Oct	Submit the P1 PDF on Canvas; tag the repository baseline v0.1.	Alexander Eng
Wed 21 Oct	Spike: confirm availability can be read without credentials (A1), measure alert delay, check free-tier limits for email and push; write findings in /docs/spikes.	Arshan Khimani (availability, alert delay); Isaam Memon (free-tier limits)
Fri 23 Oct	Send questions Q1–Q4 and Q6 to library staff and OIT; log replies in /docs/elicitation.	Lee Baillard
Sun 25 Oct	Student survey with at least 10 respondents covering Q5 and the QR-01 and FR-13 thresholds; summary in /docs/elicitation.	Mayand Dhami
Fri 30 Oct	Requirements v0.2 through change control; Q1–Q6 each resolved or restated.	Alexander Eng
Wed 4 Nov	Draft architecture (Ch. 8) and class and sequence diagrams for UC-01 and UC-02.	Arshan Khimani (architecture); Henry McGahan (class and sequence diagrams)
Fri 6 Nov	Integrate P2 document; check traceability and that every requirement ID is still stable.	Isaam Memon
Sun 8 Nov	Submit P2.	Arshan Khimani

4.3 Risks

#	Risk	Likelihood	Impact	Mitigation — action and owner
R1	The library's availability pages cannot be read by a program, or change format, so FR-03 fails (A1).	Medium	High	Arshan runs the spike by 21 Oct. If it fails he reports on 23 Oct and the team re-scopes through change control to a demonstration against recorded sample data, and tells the instructor.
R2	The library or OIT objects to automated reading and asks us to stop (A2).	Low	High	Lee asks the library on 23 Oct (Q2). QR-05 caps our request rate from the first run, and Arshan pauses reading at once if asked.
R3	NetID sign-on cannot be used by a student project (A3).	Medium	Medium	Lee asks OIT on 23 Oct (Q3). If refused, Alexander raises a change request to CON-01 by 30 Oct.
R4	Members are unavailable around midterms, leaving a task without an owner or a reviewer.	Medium	High	Isaam reviews progress against 4.2 on 25 Oct and 1 Nov; every task has a named backup recorded in its issue. If behind, FR-14 and then FR-08 are cut first through change control.
R5	Alerts arrive too late to get the room (QR-01 missed) or sit unseen (A5).	Medium	Medium	Arshan measures alert delay with scripted releases by 4 Nov. If QR-01 cannot be met, Mayand revises it on the survey evidence by 6 Nov.

4.4 Repository and tooling

Repository: https://github.com/PaperBagTrick/Study-Room-Sniper

Access: Prof. Sekir

Branches: main holds only reviewed work. Changes are made on short-lived branches named type/short-description and merged by a pull request reviewed by at least one other member.

Commits: one logical change each, and the message starts with the issue number it serves (for example "#12 add FR-04 acceptance test").

Issues: every requirement ID (FR-, QR-, CON-) has an issue with that ID in its title. A change to a baselined requirement is made through an issue of labelled change-request. Documents live in /docs, spike notes in /docs/spikes, and elicitation notes in /docs/elicitation.

4.5 Declaration of AI tool use

Tool	What it was used for	Who used it	How the output was checked
Claude (Anthropic)	Drafted the project concept, the text of sections 1–6, the requirements, use cases and plan from the course template, and wrote the code that drew the two diagrams.	Alexander Eng	Check each library fact against R2–R4 and testing each acceptance criterion for testability.
Claude (Anthropic) web search	Looked up the UTD library's room counts and booking rules used in [R2]–[R4].	Alexander Eng	opened the library pages cited as R2–R4 and compared each fact used in this document against the page text.

5 Open Questions

#	The question	Who can answer it	Needed by	What you assumed in the meantime

Q1	Does the library reservation system give programs a supported way to read room availability (feed, API or public page), and on what terms?	McDermott Library reservations or systems staff	27 Oct	Yes: Availability can be read without logging in (A1).
Q2	May an unofficial service read availability automatically at up to 20 requests per minute, and tell students about openings?	McDermott Library reservations staff	27 Oct	Yes, read-only. We pause reading at once if asked.
Q3	Can a student project use NetID for single sign-on, and what identity data (NetID, email) would it receive?	UTD OIT	30 Oct	Yes, and we receive a NetID and an email address.
Q4	What makes a free slot bookable for a particular student: the 7-day horizon [R2], the four-hour maximum and one active reservation per 24 hours [R3], rooms closing before the library does? Should alerts take a student's existing reservation into account?	Library staff, and the policy pages [R3], [R4]	30 Oct	A slot as shown free is bookable by any student; alerts ignore a student's own reservations.
Q5	Should a watch stay Active after it alerts (our current plan), and are a 3-minute alert time and a limit of 5 Active watches acceptable to students?	UTD students (survey)	25 Oct	The watch stays Active until it expires or is cancelled; 3 minutes and 5 watches.
Q6	Which accessibility standard does UTD require of a campus-facing service?	UTD OIT or a web accessibility contact	30 Oct	WCAG 2.2 Level AA [R5].

6 Baseline Declaration

We confirm that the requirements in section 3 represent our agreed baseline as of the submission date, and that every team member has read this document in full.

Team member	Signature or typed name	Date
Alexander Eng	*Alexander Eng*	10/18/2026
Arshan Khimani	Arshan Khimani	10/18/2026
Lee Baillard	Lee Baillard	10/18/2026
Mayand Dhami	Mayand Dhami	10/18/2026
Henry McGahan	Henry McGahan	10/18/2026
Isaam Memon	Isaam Memon	10/18/2026